

Angles, Polygons & Parallel Lines

Answer sheet · 28 marks total

Section A — Angle facts

A1. $180 - 118 = 62^\circ$ (1)

A2. $360 - 95 - 140 = 125^\circ$ (1)

A3. 42° (1) vertically opposite angles are equal (1)

A4. $4x + 2x + 30 = 180$ (1) $6x = 150$, $x = 25$ (1)

Section B — Triangles & quadrilaterals

B1. $180 - 48 - 76 = 56^\circ$ (1)

B2. Other base angle = 70° (1) apex angle = $180 - 70 - 70 = 40^\circ$ (1)

B3. $360 - 90 - 105 - 88 = 77^\circ$ (1)

B4. $100 + 70 + 2x = 360$ (1) $2x = 190$, $x = 95^\circ$ (1)

Section C — Parallel lines

C1. **Alternate** angles (1)

C2. 108° (corresponding angles are equal) (1)

C3. $180 - 62 = 118^\circ$ (co-interior angles sum to 180°) (1)

C4. Corresponding angles are equal, so $2x + 15 = 85$ (1) $2x = 70$ (1) $x = 35$ (1)

Section D — Polygons

D1. $(6 - 2) \times 180 = 720^\circ$ (1)

D2. $360 \div 12 = 30^\circ$ (1)

D3. Exterior = $360 \div 10 = 36^\circ$, interior = $180 - 36 = 144^\circ$ (1)

D4. Exterior angle = $180 - 150 = 30^\circ$ (1) $n = 360 \div 30 = 12$ sides (1)

Marking note: award method marks for a correctly set-up equation or calculation even if the final answer contains an arithmetic slip. Accept equivalent correct reasoning for "giving a reason" parts (e.g. naming the angle rule used). Missing units ($^\circ$) on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.